Curriculum Vitae

Shayan Hundrieser

Date of Birth	May 18^{th}, 1996	**Affiliations**	University of Göttingen Göttingen, Germany, and University of Twente Enschede, The Netherlands
Languages	German (native speaker), English (fluent)		
Nationality	German		

Work Experience

May 2025 - Present Postdoctoral Researcher – University of Twente, Enschede, The Netherlands
Research interests: Statistical Theory and Methodology for Optimal Transport
Funded by Leopoldina Postdoctoral Scholarship (05/2025 – 04/2026: University of Twente, Enschede, The Netherlands; 05/2026 – 04/2027 at Yale University, New Haven, CT, USA)

Apr 2020 - Present Researcher – University of Göttingen, Göttingen, Germany
Research interests: Statistical Analysis of Optimal Transport and their Applications, Super-resolution Microscopy and Deconvolution, Estimation in Finite Mixture Models, Statistical Methodology with Fréchet Means on Metric Spaces

Education

Apr 2020 - Feb 2024 Ph.D. studies in Mathematical Sciences – University of Göttingen, Germany
Ph.D. Thesis: *"Statistical Optimal Transport and its Entropic Regularization: Compared and Contrasted"*, *Grade: Summa Cum Laude* (supervisor: Prof. Dr. Axel Munk), awarded with two dissertation prizes

Oct 2017 - Mar 2020 M.Sc. in Mathematics, Minor in Economics – University of Göttingen, Germany
Grade: Very good (1.0) *with distinction*
Specialization: Mathematical Statistics, Statistical Data Analysis, Optimization

Oct 2014 - Sep 2017 B.Sc. in Mathematics, Minor in Economics – University of Göttingen, Germany
Grade: Very good (1.1) *with distinction*
Specialization: Mathematical Statistics, Statistics on Non-Euclidean Spaces

Awards, Prizes, and Scholarships

2025 - 2027 Leopoldina National Academy of Science Postdoctoral Scholarship
Project title: "Towards high-dimensional statistical optimal transport with sparse PCA and neural networks" — *15 – 20 fellows per year within Germany, Austria and Switzerland*
Full funding for one year at University of Twente, Enschede, Netherlands and one year at Yale University, New Haven, CT, USA.

2025 - 2027 Walter-Benjamin Postdoctoral Scholarship by German Research Foundation (DFG, Deutsche Forschungsgemeinschaft) – Declined due to acceptance of Leopoldina Postdoctoral Scholarship.

2025 Dissertation Prize of the University Association of Göttingen (registered association)
(Dissertationspreis des Universitätsbundes Göttingen e.V.)
honoring the best dissertation of the University of Göttingen

2025 Dissertation Prize by the Probability and Statistics Group of the German Mathematical Society
(Dissertationspreis der Fachgruppe Stochastik der Deutschen Mathematiker-Vereinigung)
honoring the best dissertation in Germany in the field of Mathematical Statistics

2025 Junior Researcher Travel Award to attend the IMS Conference on Statistics and Data Science in Seville, Spain in December 2025

2024 Faculty award for excellent Doctoral studies in Mathematical Sciences

2020 Faculty award for an excellent Master of Science degree in Mathematics

2018 - 2020 Fellow of German Academic Scholarship Foundation (Studienstiftung d. deutschen Volkes) *Germany's most prestigious academic scholarship*

2017 Faculty award for an excellent Bachelor of Science degree in Mathematics

Publications

Preprints

1. **Hundrieser, S.**, Tuchel, P., Kong, I., and Schmidt-Hieber, J. (2025). *On the Universal Representation Property of Spiking Neural Networks*, submitted [preprint arXiv:2512.16872].

2. Groppe, M., Niemöller, L., **Hundrieser, S.**, Ventzke, D., Blob, A., Köster, S., and Munk, A. (2025). *Optimal transport based testing in factorial design*, submitted [preprint arXiv:2509.13970].

3. Struleva*, M., **Hundrieser*, S.**, Schuhmacher, D., and Munk, A. (2025). *Sharp convergence rates of empirical unbalanced optimal transport for spatio-temporal point processes*, submitted [preprint arXiv:2509.04225].

4. **Hundrieser*, S.**, Manole*, T., Litskevich, D., and Munk, A. (2025). *Local Poisson deconvolution for discrete signals* [preprint arXiv:2508.00824].

5. **Hundrieser, S.**, Eltzner, B., and Huckemann, S. (2024). *A lower bound for estimating Fréchet means*, submitted to Annals of the Institute of Statistical Mathematics, revised [preprint arXiv:2402.12290].

6. González-Sanz, A. and **Hundrieser, S.** (2023). *Weak limits for empirical entropic optimal transport: Beyond smooth costs*, submitted [preprint arXiv:2305.09745].

Peer-Reviewed Publications

7. **Hundrieser*, S.**, Heinemann*, F., Klatt, M., Struleva, M., and Munk, A. (2025). *Unbalanced Kantorovich-Rubinstein distance, plan, and barycenter on finite spaces: A statistical perspective*, Journal of Machine Learning Research 26(37), pp. 1–70.

8. Staudt, T., **Hundrieser, S.**, and Munk, A. (2025). *On the uniqueness of Kantorovich potentials*, SIAM Journal on Mathematical Analysis, 57(2) pp. 1452–1482.

9. Staudt, T. and **Hundrieser, S.** (2025). *Convergence of empirical optimal transport in unbounded settings*, Bernoulli 30(4), pp. 2846–2877.

10. Groppe, M. and **Hundrieser, S.**, (2024). *Lower complexity adaptation for empirical entropic optimal transport*, Journal of Machine Learning Research, 25(344), pp. 1–55, awarded with Student Travel Award from ICSDS 2023, Lisbon, Portugal.

11. **Hundrieser, S.**, Mordant, G., Weitkamp, C.A., and Munk, A. (2024). *Empirical optimal transport under estimated costs: Distributional limits and statistical applications*, Stochastic Processes and their Applications, 178(104462), pp. 1–45.

*Equal contribution

12. **Hundrieser, S.**, Eltzner, B., and Huckemann, S. (2024). *Finite sample smeariness of Fréchet means with application to climate*, Electronic Journal of Statistics, 18(2), pp. 3274–3309.

13. **Hundrieser, S.**, Klatt, M., Munk, A., and Staudt, T. (2024). *A unifying approach to distributional limits for empirical optimal transport*, Bernoulli, 30(4) pp. 2846–2877.

14. **Hundrieser, S.**, Staudt, T., and Munk, A. (2024). *Empirical optimal transport between different measures adapts to lower complexity*, Annales de l'Institut Henri Poincaré, Probabilités et Statistiques, 60(2), pp. 824–846.

15. **Hundrieser, S.**, Klatt, M., and Munk, A. (2024). *Limit distributions and sensitivity analysis for empirical entropic optimal transport on countable spaces*, The Annals of Applied Probability, 34(1B), pp. 1403–1468.

16. **Hundrieser, S.**, Klatt, M., and Munk, A. (2022). *The statistics of circular optimal transport*, Proceedings of Directional Statistics for Innovative Applications, pp. 57–82.

17. Eltzner, B., **Hundrieser, S.**, and Huckemann, S. (2021). *Finite sample smeariness on spheres*, Proceedings of 5th International Conference on Geometric Science of Information, pp. 12–19.

Software

1. Graphical user interface "RSNN - Recurrent Spiking Neural Networks" for simulation purposes, available on "*github.com/hundrieser/RSNN*"

2. R-package "FSS" based on publication "Finite sample smeariness of Fréchet means on the circle and the torus" available on "*github.com/hundrieser/FSS*".

3. R-package "circularOT" based on publication "The statistics of circular optimal transport" available on "*gitlab.gwdg.de/shundri/circularOT*".

Oral Presentations

Dec 2025 *Local Deconvolution of Discrete Signals under Wasserstein Loss*
IMS International Conference on Statistics and Data Science, Seville, Spain (Invited Talk)

Nov 2025 *Is Deconvolution Still Hard? – New Perspectives in the Superresolution Microscopy Era*
University of Twente, Enschede, The Netherlands (Invited Talk)

Nov 2025 *Is Deconvolution Still Hard? – New Perspectives in the Superresolution Microscopy Era*
Technical University Delft, Delft, The Netherlands (Invited Talk)

May 2025 *New frontiers in statistical optimal transport*
Workshop on the Statistical Theory of Neural Networks (Invited Talk)
Egmond aan Zee, The Netherlands

Mar 2025 *Statistical aspects of optimal transport: Regularization, estimation, and applications*
German Probability and Statistics Days 2025, Dresden, Germany (Invited Talk)

Jan 2025 *Statistical unbalanced optimal transport*
Heidelberg-Paris Workshop on Mathematical Statistics (Invited Talk)
Heidelberg, Germany

Sep 2024 *Empirical optimal transport: Convergence rates and lower complexity adaptation*
Tata Institute of Fundamental Research, Bangalore, India (Invited Talk)

Aug 2024 *Statistical optimal transport and its entropic regularization: Compared and contrasted*
Bernoulli World Congress, Bochum, Germany (Invited Talk)

Jul 2024 *Low intrinsic dimensionality is all you need*
Workshop on Statistics & Learning Theory in the Era of AI (Invited Talk)
Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach-Walke, Germany

Mar 2024 *Empirical optimal transport: Convergence rates and lower complexity adaptation*
University of Twente, Enschede, Netherlands (Invited Talk)

Jan 2024 *Optimal transport and its entropic penalization: A statistical exploration*
University of Cambridge, Cambridge, United Kingdom (Invited Talk)

Dec 2023 *A unifying approach to distributional limits for empirical optimal transport*
IMS International Conference on Statistics and Data Science, Lisbon, Portugal

Oct 2023 *Empirical optimal transport: Convergence rates and lower complexity adaptation*
University of Cambridge, Cambridge, United Kingdom (Invited Talk)

Jul 2023 *Empirical optimal transport: Convergence rates and lower complexity adaptation*
Carnegie Mellon University, Pittsburgh, Pennsylvania, USA (Invited Talk)

Aug 2022 *Empirical optimal transport and the lower complexity adaptation principle*
7th Annual Research Training Group 2088 Workshop, Goslar, Germany

Sep 2021 *The statistics of circular optimal transport*
German Probability and Statistics Days, Mannheim, Germany

May 2021 *Entropic optimal transport on countable spaces: Statistical theory and asymptotics*
Entropy 2021: The Scientific Tool of the 21st Century, Lisbon, Portugal

Teaching Experience

Fall 2024 Assistant for lecture "Statistical foundations of data science"

Spring 2024 Seminar on "Empirical processes"

Fall 2023 Assistant for lecture "Discrete stochastics"

Fall 2023 Seminar on "Optimal transport: Foundations, computation, and statistics"

Fall 2022 Assistant for lecture "Statistical foundations of data science"

Fall 2020 Assistant for lecture "Introduction to spatial stochastics"

Referee Service

I was involved in referee work for the following (peer-reviewed) journals:

- The Annals of Statistics
- Bernoulli Journal
- IEEE on Information Theory
- Annales de l'Institut Henri Poincaré, Probabilités et Statistiques
- The Electronic Journal of Statistics
- Statistical Science